

Dexter Fernandes

SLAM Computer Vision Engineer

Fluent in English, Hindi, Konkani

Bristol, EN | <https://www.linkedin.com/in/dexterferns/> | +44 7436007369 | dexterfernandes11@gmail.com

PERSONAL PROFILE

Computer Vision Engineer with over 3 years of expertise in developing low-latency vision-based solutions for safety and inspection for Tier 1 companies, including a construction company in Texas, USA and a UK oil and gas company in Muscat, Oman. Successfully collaborated with multinational clients to address technical challenges and achieve significant milestones within a 4-month timeframe.

KEY ACHIEVEMENTS

- Achieved a 3.6x speedup in vision model inference speed on the same inference hardware.
- Reduced model training time from 2.5 days to 4 hours on the same training hardware.
- Eliminated a year-old segmentation fault in a massive monolithic application.
- Built a MLOps pipeline that automated the training of 300+ models within two days.

AREAS OF EXPERTISE

Deep Learning	Computer Vision	Time-series Analysis
Hyperparameter Tuning	Version Control	Containerisation
Leadership and Management	Client Interaction	MLOps
Distributed Event Data-Streaming	Project Management	SLAM

WORK EXPERIENCE

Esbaar <i>AI Engineer</i>	Muscat, Oman <i>January 2023 – August 2023</i>
-------------------------------------	--

Z-Alert

- Developed custom time-series models in Keras and PyTorch to classify 15 oil rig activities, managing and analysing 3.6 million data points from numerous sensors.
- Streamlined hyperparameter tuning for time-series analysis models through MLOps integration, orchestrating the training of approximately 500 time-series models within a 48-hour period – improving F1-score by 39%.
- Led the team that adapted YOLOv5 to detect rig workers and heavy equipment, fostering a safer work environment, improving model F1-score by 17% through model maintenance, iteration, and drift management.
- Collaborated with multinational clients to address technical challenges, achieving key milestones and delivering solutions within 4 months, while gaining valuable insights into cross-cultural communication and project management.
- Leveraged Apache Kafka with Docker containers to re-factor a monolithic application into a microservices architecture, reducing software update intervals and improving customer satisfaction.

LivNSense Technologies Pvt. Ltd.
Lead Software Engineer - AI

Bengaluru, India
May 2020 – December 2022

VICAS – Safety-based Anti-Collision Early Detection System

- Managed a team of 2-4 persons that contributed to every part of the pipeline, which included curating the training and testing datasets, managing model life cycle in production, and tackling edge cases.
- Overhauled the labelling platform and repurposed older production models for auto-labelling, increasing annotator output from 600 to 3000 images per day and maximising resource efficiency.
- Surpassed the existing vision solution by revamping the dataset, re-formulating the problem statement, and reducing inference time from 250 ms to 90 ms on the same edge hardware.
- Systematically resolved significant model drift in a single day by constructing and training on a synthetic dataset mimicking production changes, reducing false positives by 50% and reassuring clients until raw data was procured.
- Represented the company as a consultant abroad, understanding client requirements and delivering a pilot solution with real-time analytics within two weeks. – Workplace Hazard Detection System

EDUCATION

University of Bristol
MSc in Robotics

Bristol, England
September 2023 – September 2024

Visvesvaraya National Institute of Technology
B. Tech in Electrical and Electronics Engineering

Nagpur, India
July 2016 – May 2020

SOFTWARE STACK

Python	C++	OpenCV	Intel OpenVINO
Sci-kit Learn	MLFlow	Optuna	Github
Docker	Jira	CVAT	Matplotlib

CERTIFICATIONS

Deep Learning Specialisation – 5-course specialisation at deeplearning.ai

AWARDS

Received the **Best Speaker award** on three occasions from ABC Toastmasters club in Muscat, Oman.